Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Deep Learning Surrogate Modeling and Design Optimization of Battery Thermal Management Systems**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery (√), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Advanced Mobility		Artificial Intelligence	
	Medium Category	Small Category	Medium Category	Small Category
	Secondary Batteries	Thermal Management	Deep Learning	Convolutional Neural Networks (CNN)
Concept of Proposed Technology	Pioneering surrogate modeling with DeepEDH-CNN: Deep encoder–decoder hierarchical (DeepEDH) convolutional neural network (CNN) modelling methodology for rapid design optimization of battery thermal management systems.			
Rationale for Joint R&D Project	The application of the DeepEDH-CNN methodology to battery systems requires in-depth knowledge of the battery system, including battery cell specifications, packaging components, thermophysical properties, thermal management system, and performance data. This is proprietary information accessible only through a collaboration with the developer of the commercial battery system or thermal management system.			
Key Words	1. Secondar battery thermal management			

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	2. Battery heat generation and transport 3. Conjugate heat transfer systems 4. Deep learning surrogate modeling 5. Design optimization

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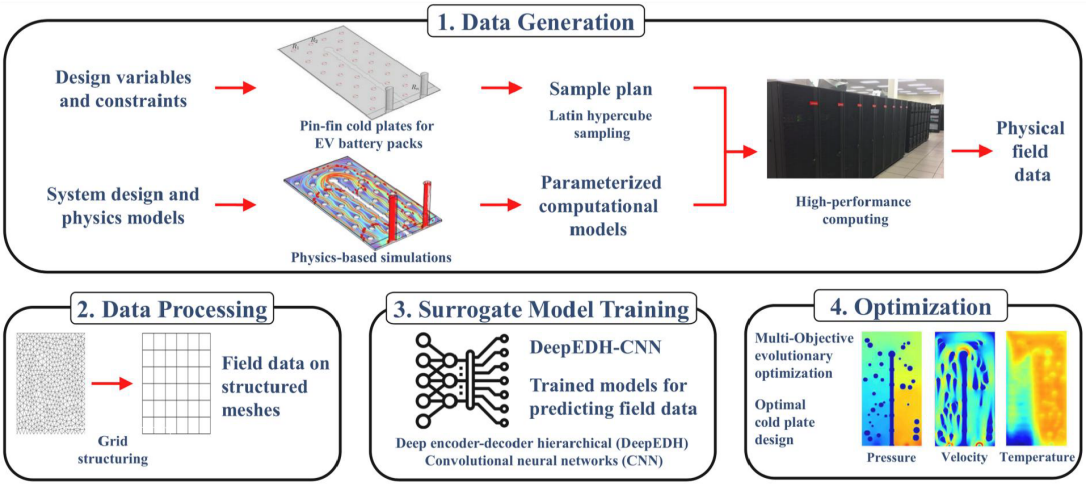
3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2)	End of the Project	Level (6)
R&D Objectives	Developed a generalized surrogate modeling DeepEDH-CNN methodology for the reconstruction of physical fields for conjugate heat transfer systems, leveraging hierarchical physics-based models of battery thermal management systems to generate training data. Demonstrate the capabilities of the DeepEDH-CNN methodology as an effective design optimization methodology for battery thermal management systems, achieving remarkable computational savings.			
R&D Target Specifications	Achieve optimal battery thermal management system designs in a few hours using the DeepEDH-CNN surrogate modeling methodology, compared to months of computational time if conventional physics-based models were employed.			
R&D Plan	The Figure below outlines the proposed DeepEDH-CNN methodology, which involves the following steps: (1) generating sample points with Latin Hypercube sampling and numerically simulating temperature, velocity, and pressure fields for each sample point; (2) converting unstructured simulated results to structured image-like results, as required by convolutional neural network surrogate models; (3) training the surrogate model with the DeepEDH-CNN methodology, employing dense blocks in a fully convolutional encoder–decoder structure specifically designed for the reconstruction of physical fields for conjugate heat transfer systems; and (4) replacing the computationally expensive physics-based model with the surrogate model in the optimization process.  			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	1) Conduct thermal characterization activities to predict heat generation and thermophysical properties of secondary batteries. 2) Build computationally efficient and validated hierarchical thermal models of the target battery system. 3) Generate data with the thermal models to later train a surrogate model. 4) Develop the surrogate model with the DeepEDH-CNN methodology. 5) Apply a multi-objective design optimization framework with an evolutionary optimizer to determine the optimal designs of battery thermal management systems that maximize the lifespan of battery systems.
Expected R&R of Korean SME Partner	1) Provide cell specifications, packaging components, thermophysical properties, thermal management system, and performance data. 2) Support and validate the development of the surrogate model. 3) Prototype and evaluate the performance of the optimal thermal management designs in industrial testbeds.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The deep learning DeepEDH-CNN methodology will overcome the challenges posed by computationally expensive physics-based simulations by leveraging lower-cost physics-informed data-based surrogate models. Design optimization with DeepEDH-CNN will enable faster, more thorough optimization processes and a deeper exploration of the optimization space of battery thermal management systems. The DeepEDH-CNN methodology is expected to accelerate the design process of next-generation battery thermal management systems from months to just a few hours.

6. References

[1] Ebbs-Picken, T., Da Silva, C.M., and Amon, C.H., Hierarchical Modeling and Surrogate-model-based Design Optimization Framework for Cold Plates used in Battery Thermal Management Systems, Applied Thermal Engineering, 253, 2024.

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[7] Da Silva, C.M., Akula, R., and Amon, C.H., Challenges and Opportunities in Hierarchical Multi-Length-Scale Thermal Modeling of Electric Vehicle Battery Systems. ASME Journal of Heat and Mass Transfer, pp.1-35, 2025.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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